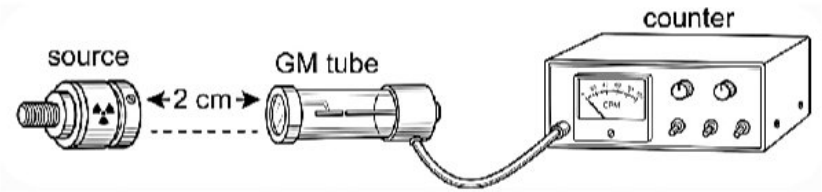
29 An experiment is set up to determine the nature of the radiation emitted by a radioactive isotope and to estimate its half-life. The source is placed 2 cm away from a Geiger-Muller (GM) tube connected to a counter. The count-rate is measured after each duration, *t*, has elapsed.



The following data is obtained:

t / hours	Count-rate with source / min^{-1}	Count-rate with source (behind a thin sheet of paper) / min^{-1}	Count-rate with source (behind a 5 mm aluminum sheet) / min^{-1}
0	3250	3247	31
24	430	431	29

Which row correctly identifies the primary type of emission and the half-life of the source?

	type of emission	half-life / hours
A	alpha	8
B	alpha	12
C	beta	8
D	beta	12